UPSKILL SPRINT · WALL REFERENCE · LEAN SIX SIGMA & QUALITY

Which Excel function do I need?

Start with your question. Pick the branch, then the sub-question. Every one of Excel's 111 statistical functions, 21 helpers and 27 tools is an answer below.

What do you need to know?

01 Describe what I have · 02 How likely is it? · 03 How sure am I? · 04 Is the difference real? · 05 Do they move together? · 06 Can I model Y from X? · 07 What comes next? · 08 What shape is it in? · 09 What settings hit the target? · 10 Want the full output table? · 11 Old workbook, old name? · 12 What can't Excel do?

01 · DESCRIPTIVE STATISTICS 36
Describe what I have

— WHERE IS THE CENTRE?

- **AVERAGE**(number1, [number2], ...)
  Default centre of a data set
- **AVERAGEA**(value1, [value2], ...)
  Only when logical values must count
- **AVERAGEIF**(range, criteria, [average_range])
  Mean for one shift, grade or line
- **AVERAGEIFS**(average_range, criteria_range1, criteria1, ...)
  Mean by heat and orientation together
- **MEDIAN**(number1, [number2], ...)
  Skewed data such as delays and cycle times
- **MODE.SNGL**(number1, [number2], ...)
  Discrete readings with repeats
- **MODE.MULT**(number1, [number2], ...)
  Data with two or more peaks
- **GEOMEAN**(number1, [number2], ...)
  Growth rates, ratios, log-normal data
- **HARMEAN**(number1, [number2], ...)
  Averaging rates such as units per hour
- **TRIMMEAN**(array, percent)
  A robust mean when outliers are present

— HOW SPREAD OUT IS IT?

- **STDEV.S**(number1, [number2], ...)
  Almost every quality calculation on sample data
- **STDEV.P**(number1, [number2], ...)
  Only when you hold the whole population
- **STDEVA**(value1, [value2], ...)
  Rarely; text becomes 0 and distorts results
- **STDEVPA**(value1, [value2], ...)
  Rarely; same caution as STDEVA
- **VAR.S**(number1, [number2], ...)
  ANOVA, F-tests and pooled variance
- **VAR.P**(number1, [number2], ...)
  Complete populations only
- **VARA**(value1, [value2], ...)
  Rarely
- **VARPA**(value1, [value2], ...)
  Rarely
- **AVEDEV**(number1, [number2], ...)
  A spread summary that resists outliers
- **DEVSQ**(number1, [number2], ...)
  Building ANOVA sums of squares by hand

— WHAT SHAPE IS IT?

- **SKEW**(number1, [number2], ...)
  Quick normality and symmetry screen
- **SKEW.P**(number1, [number2], ...)
  Skewness of a complete population
- **KURT**(number1, [number2], ...)
  Screening for heavy or light tails

— WHAT ARE THE EXTREMES?

- **MAX**(number1, [number2], ...)
  Range and worst-case checks
- **MAXA**(value1, [value2], ...)
  Rarely
- **MAXIFS**(max_range, criteria_range1, criteria1, ...)
  Worst reading per grade or line
- **MIN**(number1, [number2], ...)
  Range and minimum-spec checks
- **MINA**(value1, [value2], ...)
  Rarely
- **MINIFS**(min_range, criteria_range1, criteria1, ...)
  Lowest toughness per heat
- **LARGE**(array, k)
  Top-k values and order statistics
- **SMALL**(array, k)
  Bottom-k values and order statistics

— HOW MANY ARE THERE?

- **COUNT**(value1, [value2], ...)
  The sample size for every other formula
- **COUNTA**(value1, [value2], ...)
  Checking for text entries in numeric columns
- **COUNTBLANK**(range)
  Missing-data audit
- **COUNTIF**(range, criteria)
  Defects beyond a spec limit
- **COUNTIFS**(criteria_range1, criteria1, ...)
  Contingency tables and DPU by category

02 · PROBABILITY DISTRIBUTIONS 41
How likely is it?

— NORMAL (BELL) CURVE?

- **NORM.DIST**(x, mean, standard_dev, cumulative)
  Share of output below a limit, ppm out of spec
- **NORM.INV**(probability, mean, standard_dev)
  Percentile limits; simulation with RAND()
- **NORM.S.DIST**(z, cumulative)
  p-values for z statistics
- **NORM.S.INV**(probability)
  Critical z, sample size, sigma level
- **GAUSS**(z)
  Quick area from the centre
- **PHI**(x)
  Plotting the bell curve

— STUDENT T?

- **T.DIST**(x, deg_freedom, cumulative)
  Lower-tailed t tests
- **T.DIST.RT**(x, deg_freedom)
  Upper-tailed t tests
- **T.DIST.2T**(x, deg_freedom)
  Two-sided p-value for a t statistic
- **T.INV**(probability, deg_freedom)
  One-sided critical values; negative for small p
- **T.INV.2T**(probability, deg_freedom)
  Confidence intervals and two-sided tests

— CHI-SQUARE OR F?

- **CHISQ.DIST**(x, deg_freedom, cumulative)
  Lower tail of a variance test
- **CHISQ.DIST.RT**(x, deg_freedom)
  p-value for chi-square tests with any df
- **CHISQ.INV**(probability, deg_freedom)
  Upper limit of a variance confidence interval
- **CHISQ.INV.RT**(probability, deg_freedom)
  Chi-square table lookups; variance intervals
- **F.DIST**(x, deg_freedom1, deg_freedom2, cumulative)
  Lower tail of a variance-ratio test
- **F.DIST.RT**(x, deg_freedom1, deg_freedom2)
  ANOVA and regression p-values
- **F.INV**(probability, deg_freedom1, deg_freedom2)
  Lower limit of a variance-ratio interval
- **F.INV.RT**(probability, deg_freedom1, deg_freedom2)
  F tables for ANOVA

— COUNTING DEFECTS OR DEFECTIVES?

- **BINOM.DIST**(number_s, trials, probability_s, cumulative)
  Defectives in a sample, pass or fail data
- **BINOM.DIST.RANGE**(trials, probability_s, number_s, [number_s2])
  Acceptance-sampling ranges in one step
- **BINOM.INV**(trials, probability_s, alpha)
  Acceptance numbers and exact critical counts
- **NEGBINOM.DIST**(number_f, number_s, probability_s, cumulative)
  Inspections until a set number of finds
- **HYPGEOM.DIST**(sample_s, number_sample, population_s, number_pop, cumulative)
  Sampling small lots without replacement
- **POISSON.DIST**(x, mean, cumulative)
  Defects per unit, c- and u-chart probabilities
- **PERMUT**(number, number_chosen)
  Counting sequences and orders
- **PERMUTATIONA**(number, number_chosen)
  Codes and passwords style counting
- **PROB**(x_range, prob_range, [lower_limit], [upper_limit])
  Custom discrete distributions
- **COMBIN**(number, number_chosen) HELPER
  Counting samples where order does not matter
- **COMBINA**(number, number_chosen) HELPER
  Counting with replacement

— TIME, LIFE OR SKEWED DATA?

- **EXPON.DIST**(x, lambda, cumulative)
  Time between failures, constant failure rate
- **WEIBULL.DIST**(x, alpha, beta, cumulative)
  Life data and wear-out reliability
- **LOGNORM.DIST**(x, mean, standard_dev, cumulative)
  Repair times, particle sizes, skewed data
- **LOGNORM.INV**(probability, mean, standard_dev)
  Percentile limits for skewed data
- **GAMMA.DIST**(x, alpha, beta, cumulative)
  Waiting times for several events
- **GAMMA.INV**(probability, alpha, beta)
  Percentiles of gamma-shaped data
- **GAMMA**(number)
  Building other distributions by hand
- **GAMMALN**(x)
  Large factorials without overflow
- **GAMMALN.PRECISE**(x)
  Same use, newer algorithm
- **BETA.DIST**(x, alpha, beta, cumulative, [A], [B])
  PERT task times, proportions
- **BETA.INV**(probability, alpha, beta, [A], [B])
  Exact binomial confidence limits

03 · STATISTICAL INFERENCE 2
How sure am I?

- **CONFIDENCE.T**(alpha, standard_dev, size)
  Confidence interval for a mean, sigma unknown
- **CONFIDENCE.NORM**(alpha, standard_dev, size)
  Confidence interval for a mean, sigma known

04 · HYPOTHESIS TESTS 4
Is the difference real?

- **T.TEST**(array1, array2, tails, type)
  Comparing two means
- **Z.TEST**(array, x, [sigma])
  Mean against a target when sigma is known
- **F.TEST**(array1, array2)
  Checking variances before choosing T.TEST type
- **CHISQ.TEST**(actual_range, expected_range)
  Counts by category; contingency tables

05 · CORRELATION 7
Do they move together?

- **CORREL**(array1, array2)
  Strength and direction of a straight-line link
- **PEARSON**(array1, array2)
  Legacy sheets; identical result
- **RSQ**(known_y's, known_x's)
  Share of variation explained by one x
- **COVARIANCE.S**(array1, array2)
  Building r by hand on sample data
- **COVARIANCE.P**(array1, array2)
  Matches the ToolPak Covariance tool
- **FISHER**(x)
  Confidence interval on r; comparing two r values
- **FISHERINV**(y)
  Second half of the r interval

06 · REGRESSION 7
Can I model Y from X?

- **LINEST**(known_y's, [known_x's], [const], [stats])
  Simple or multiple regression with inference
- **SLOPE**(known_y's, known_x's)
  Change in y per unit of x
- **INTERCEPT**(known_y's, known_x's)
  y when x is zero
- **STEYX**(known_y's, known_x's)
  Width of prediction intervals
- **TREND**(known_y's, [known_x's], [new_x's], [const])
  Fitted values, residuals, predictions
- **LOGEST**(known_y's, [known_x's], [const], [stats])
  Growth or decay curves
- **GROWTH**(known_y's, [known_x's], [new_x's], [const])
  Predictions from an exponential model

07 · FORECASTING 6
What comes next?

- **FORECAST.LINEAR**(x, known_y's, known_x's)
  Straight-line trend forecasts
- **FORECAST**(x, known_y's, known_x's)
  Older workbooks only
- **FORECAST.ETS**(target_date, values, timeline, [seasonality], [data_completion], [aggregation])
  Time series with trend and seasonality
- **FORECAST.ETS.CONFINT**(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation])
  Error bands around ETS forecasts
- **FORECAST.ETS.SEASONALITY**(values, timeline, [data_completion], [aggregation])
  Confirming the cycle before trusting a forecast
- **FORECAST.ETS.STAT**(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation])
  Judging forecast accuracy

08 · EXPLORATORY DATA ANALYSIS 18
What shape is it in?

— WHERE DOES A VALUE SIT?

- **FREQUENCY**(data_array, bins_array)
  Histograms and goodness-of-fit counts
- **QUARTILE.EXC**(array, quart)
  Box plots that match Minitab
- **QUARTILE.INC**(array, quart)
  Matching older Excel workbooks
- **PERCENTILE.EXC**(array, k)
  P5 or P95 limits that match Minitab
- **PERCENTILE.INC**(array, k)
  Percentiles that include 0 and 1
- **PERCENTRANK.EXC**(array, x, [significance])
  Where one value sits in the data
- **PERCENTRANK.INC**(array, x, [significance])
  Same, inclusive method
- **RANK.AVG**(number, ref, [order])
  Normal probability plots and rank tests
- **RANK.EQ**(number, ref, [order])
  League tables and Pareto ordering
- **STANDARDIZE**(x, mean, standard_dev)
  Flagging points beyond three standard deviations

— HOW DO I SLICE AND SUMMARISE?

- **GROUPBY**(row_fields, values, function, [field_headers], ...) HELPER
  Stratify by heat, shift, line or supplier
- **PIVOTBY**(row_fields, col_fields, values, function, ...) HELPER
  Side-by-side comparisons such as LPA vs TWA
- **PERCENTOF**(data_subset, data_all) HELPER
  Pareto-style contributions
- **FILTER**(array, include, [if_empty]) HELPER
  Slicing a subgroup without copying
- **UNIQUE**(array, [by_col], [exactly_once]) HELPER
  Listing categories before grouping
- **SORT**(array, [sort_index], [sort_order], [by_col]) HELPER
  Ordering data for plots and ranks
- **SUBTOTAL**(function_num, ref1, [ref2], ...) HELPER
  Summaries that follow a filter
- **AGGREGATE**(function_num, options, ref1, [ref2], ...) HELPER
  Robust summaries on messy data

09 · WHAT-IF & OPTIMISATION 18
What settings hit the target?

— WHICH INPUT HITS MY TARGET?

- **Goal Seek** · Data › What-If Analysis › Goal Seek
  Which sigma gives Cpk 1.33?
- **Data Table (one variable)** · Data › What-If Analysis › Data Table (column input cell)
  Sensitivity of cost or yield to one input
- **Data Table (two variables)** · Data › What-If Analysis › Data Table (row and column input cells)
  Cpk map across mean and sigma
- **Scenario Manager** · Data › What-If Analysis › Scenario Manager
  Best, likely and worst cases
- **Solver (Simplex LP)** · Data › Solver › Select a Solving Method: Simplex LP
  Blends, product mix, scheduling
- **Solver (GRG Nonlinear)** · Data › Solver › Select a Solving Method: GRG Nonlinear
  Response optimization, curve fitting
- **Solver (Evolutionary)** · Data › Solver › Select a Solving Method: Evolutionary
  Models with IF, LOOKUP or rounding
- **Forecast Sheet** · Data › Forecast › Forecast Sheet
  One-click seasonal forecast

— HOW DO I SIMULATE IT?

- **RAND**() HELPER
  Driving simulations through .INV functions
- **RANDARRAY**([rows], [columns], [min], [max], [integer]) HELPER
  Thousands of draws in one formula
- **RANDBETWEEN**(bottom, top) HELPER
  Random sampling of rows
- **SEQUENCE**(rows, [columns], [start], [step]) HELPER
  Input grids for sensitivity tables
- **LET**(name1, name_value1, calculation_or_name2, ...) HELPER
  Readable one-cell simulations
- **LAMBDA**([parameter1, ...], calculation) HELPER
  Custom test statistics and models
- **MAP**(array1, lambda_or_array) HELPER
  Formula-based data tables that recalculate fast

— WHAT BUILDS THE MODEL?

- **SUMPRODUCT**(array1, [array2], ...) HELPER
  Objective and constraint cells for Solver
- **MMULT**(array1, array2) HELPER
  Regression by matrices; linear models
- **MINVERSE**(array) HELPER
  Normal equations for regression by hand

10 · ANALYSIS TOOLPAK 19
Want the full output table?

— COMPARE GROUPS?

- **Anova: Single Factor** · Data › Data Analysis › Anova: Single Factor
  Three or more group means
- **Anova: Two-Factor With Replication** · Data › Data Analysis › Anova: Two-Factor With Replication
  Two factors with repeats
- **Anova: Two-Factor Without Replication** · Data › Data Analysis › Anova: Two-Factor Without Replication
  Randomized block designs
- **t-Test: Paired Two Sample for Means** · Data › Data Analysis › t-Test: Paired Two Sample for Means
  Before and after on the same units
- **t-Test: Two-Sample Assuming Equal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances
  Independent groups, equal spread
- **t-Test: Two-Sample Assuming Unequal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances
  Independent groups, unequal spread
- **z-Test: Two Sample for Means** · Data › Data Analysis › z-Test: Two Sample for Means
  Two means with known variances
- **F-Test Two-Sample for Variances** · Data › Data Analysis › F-Test Two-Sample for Variances
  Comparing two variances

— DESCRIBE OR RELATE?

- **Descriptive Statistics** · Data › Data Analysis › Descriptive Statistics
  One-click summary
- **Regression** · Data › Data Analysis › Regression
  Full regression output
- **Correlation** · Data › Data Analysis › Correlation
  Several variables at once
- **Covariance** · Data › Data Analysis › Covariance
  Several variables at once
- **Histogram** · Data › Data Analysis › Histogram
  Quick frequency table
- **Rank and Percentile** · Data › Data Analysis › Rank and Percentile
  Relative standing table

— SIMULATE OR SMOOTH?

- **Random Number Generation** · Data › Data Analysis › Random Number Generation
  Static simulated data
- **Sampling** · Data › Data Analysis › Sampling
  Drawing an audit sample
- **Moving Average** · Data › Data Analysis › Moving Average
  Trend smoothing
- **Exponential Smoothing** · Data › Data Analysis › Exponential Smoothing
  Level-only forecasts
- **Fourier Analysis** · Data › Data Analysis › Fourier Analysis
  Vibration and periodic signals

11 · LEGACY NAMES 39
Old workbook, old name?

BETADIST → BETA.DIST
BETAINV → BETA.INV
BINOMDIST → BINOM.DIST
CHIDIST → CHISQ.DIST.RT
CHIINV → CHISQ.INV.RT
CHITEST → CHISQ.TEST
CONFIDENCE → CONFIDENCE.NORM
COVAR → COVARIANCE.P
CRITBINOM → BINOM.INV
EXPONDIST → EXPON.DIST
FDIST → F.DIST.RT
FINV → F.INV.RT
FORECAST → FORECAST.LINEAR
FTEST → F.TEST
GAMMADIST → GAMMA.DIST
GAMMAINV → GAMMA.INV
HYPGEOMDIST → HYPGEOM.DIST
LOGINV → LOGNORM.INV
LOGNORMDIST → LOGNORM.DIST
MODE → MODE.SNGL
NEGBINOMDIST → NEGBINOM.DIST
NORMDIST → NORM.DIST
NORMINV → NORM.INV
NORMSDIST → NORM.S.DIST
NORMSINV → NORM.S.INV
PERCENTILE → PERCENTILE.INC
PERCENTRANK → PERCENTRANK.INC
POISSON → POISSON.DIST
QUARTILE → QUARTILE.INC
RANK → RANK.EQ
STDEV → STDEV.S
STDEVP → STDEV.P
TDIST → T.DIST.RT or T.DIST.2T
TINV → T.INV.2T
TTEST → T.TEST
VAR → VAR.S
VARP → VAR.P
WEIBULL → WEIBULL.DIST
ZTEST → Z.TEST

12 · BEYOND EXCEL 12
What can't Excel do?

- **Normality test (Anderson-Darling, Shapiro-Wilk)**
  In Excel: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT · Python: stats.anderson, stats.shapiro
- **One- and two-proportion tests**
  In Excel: z statistic with NORM.S.DIST; exact test with BINOM.DIST · Python: proportions_ztest
- **Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)**
  In Excel: Sign test with BINOM.DIST; rank sums with RANK.AVG · Python: stats.wilcoxon, mannwhitneyu, kruskal
- **Equal variances, three or more groups (Levene, Bartlett)**
  In Excel: One-way ANOVA on absolute deviations from each group median · Python: stats.levene, stats.bartlett
- **Post-hoc comparisons (Tukey)**
  In Excel: Fisher LSD with T.INV.2T; no studentized range in Excel · Python: pairwise_tukeyhsd
- **Unbalanced or multi-factor ANOVA and DOE analysis**
  In Excel: LINEST with coded columns for two-level factorials · Python: statsmodels ols with anova_lm
- **Power and sample size**
  In Excel: Sample-size formula with NORM.S.INV · Python: statsmodels.stats.power
- **Process capability**
  In Excel: Cpk from AVERAGE and STDEV.S, or R-bar over d2 · Python: numpy formulas
- **Control charts**
  In Excel: Limits from AVERAGE and constants, drawn as a line chart · Python: matplotlib chart in the cell
- **Gage R&R**
  In Excel: ANOVA method built from DEVSQ · Python: statsmodels random-effects ANOVA
- **Weibull fitting**
  In Excel: LINEST on the linearized Weibull plot · Python: stats.weibull_min.fit
- **Logistic or Poisson regression**
  In Excel: Solver maximizing the log-likelihood · Python: statsmodels Logit or GLM
- **PY**(python_code, return_type) HELPER
  scipy.stats and statsmodels for tests Excel lacks

Rule of thumb: p below α (usually 0.05) means the difference is real. Check normality and equal variance first; Excel won't warn you.